

Up to ten years of visual acuity outcomes in fellow eyes post-uveal melanoma treatment with iodine-125 radiotherapy, transpupillary thermotherapy, and proton beam therapy

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Abstract

Purpose: This study aimed to assess long-term visual acuity outcomes and multiple secondary factors in fellow eyes of uveal melanoma patients treated with either iodine-125 (¹²⁵I) plaque radiotherapy (PRT), transpupillary thermotherapy (TTT), or proton beam therapy (PBT).

Material and methods: A retrospective chart review of 213 patients treated from 2000 to 2022 was performed, with baseline and post-treatment visual acuity data collected at follow-up time points up to 10 years, when available. Paired *t*-tests compared changes in visual acuity (VA) and intraocular pressure (IOP) over time for affected and fellow eyes. Spearman's rho test assessed correlations between melanoma thickness and VA over time, and repeated measures ANOVA determined interactions between time and comorbidities in VA.

Results: Fellow eyes VA showed no significant change up to a decade post-treatment regardless of treatment modality, except for a likely false positive improvement at 2 years post-PRT. In contrast, affected eyes had significant VA decline post-PRT and PBT, while TTT used in affected eyes demonstrated no significant change. IOP did not show any significant changes in fellow eyes. Tumor thickness and the presence of diabetes, hypertension, or coronary artery disease, did not correlate with fellow eye VA outcomes.

Conclusions: This study suggests that following uveal melanoma treatment with either ¹²⁵I PRT, TTT, or PBT, fellow eyes remain stable regarding VA and other outcomes up to 10 years. This provides important information for treatment choice in patients with ocular melanoma, especially as VA can decline in affected eye post-treatment, leading to reliance on contralateral eye.

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Purpose

Melanoma is characterized by malignant transformation of melanocytes that can occur in many areas of the body, including the skin, GI tract, and eye. Uveal melanoma (UM) arises from melanocytes in the eye's middle layer (the uvea), which includes the choroid, ciliary body, and iris [1]. Although rare, UM is the most common primary intraocular neoplasm in adults [2–4]. Moreover, UM has been shown in the U.S. to be more common in men

than women, and in Whites than patients of other races [3]. Its incidence varies by region globally, with estimates of 2–7.9 cases per million person-years in North America [4]. UM, like other melanomas, has a high tendency to metastasize. Even though there have been many UM diagnostic and treatment improvements, data suggests that metastatic spread is often established early in the tumor life, and there are no effective methods to prevent metastases. These outcomes vary based on patient-related factors, such as age, and tumor-related factors, such as

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lesion size [4, 5]. Early diagnosis and prompt, effective treatment are thus critical to optimize visual and health outcomes of UM patients.

Some common treatment options for UM include radiation therapy with plaque radiotherapy (PRT) or proton beam therapy (PBT); rare cases are treated with laser therapy, including transpupillary thermotherapy (TTT), or surgical treatment, such as enucleation or local resection. The decision of treatment choice is multifactorial; some factors considered include tumor characteristics, such as size, location, and stage, in addition to treatment availability and patient medical history [6-8]. Over the last decades, PRT has become the most common treatment modality for uveal melanoma to achieve intraocular tumor control [8, 9]. Well cited complications associated with UM treatments, such as PRT, include optic neuropathy, radiation retinopathy, maculopathy, neovascular glaucoma, increased incidence of other intraocular disease, i.e., cataracts, and enucleation in selected cases [8-11]. Furthermore, it has been well-established that affected eyes undergoing uveal melanoma treatment with radiation often suffer from deteriorating visual acuity (VA) due to radiation-induced morbidity [12]. A systematic review by Tseng *et al.* found 5 studies showing that the proportion of patients with VA $\leq$ 20/40 had less than 50% of preserving this level of VA at last follow-up post-PRT; some factors associated with poor long-term VA following this treatment included radiation dose and tumor location near the optic nerve [9]. Multiple other studies have also observed a decrease in VA values using iodine-125 (^{125}I) PRT, from baseline up to multiple years intervals, to be statistically significant [10, 11, 13-17]. Some additional factors were predictive of final VA 20/200 or less in affected eyes, including radiation papillopathy, radiation cataract, plaque size, and final tumor thickness [11]. Similarly, PRT was shown to cause significant visual loss, presumably more than plaque therapy. Tseng *et al.*'s review of 7 studies reported that less than 50% of affected eyes were able to preserve VA $\leq$ 20/200 at follow-up [9]. Some independent negative predictors of VA preservation using PRT treatment included older age, tumor size, tumor site, and radiation dose [8, 18].

To date, few studies have focused primarily on recent fellow eye outcomes, such as long-term VA, in the context of different uveal melanoma treatment modalities. A Collaborative Ocular Melanoma Study group prospective investigation (2003) on unilateral choroidal melanoma patients, who were followed up to ten years after treatment using either ^{125}I PRT or pre-enucleation external radiation, found that mean change in VA of fellow eyes from baseline to each exam was one letter (0.2 Snellen lines) or less. The same study showed that intraocular pressure (IOP) changed by less than 1 mmHg from baseline to each examination, and both treatment arms reflected similar results [19]. No other studies have examined fellow eye outcomes thoroughly, especially contrasting PRT directly with other treatment options, including PBT and TTT.

Most recent studies examining VA of fellow eyes reported secondary outcomes after a specific uveal melanoma treatment for durations shorter than 10 years [20, 21]. Other research used non-irradiated fellow eyes as con-

trols for comparison with affected eyes, showing BCVA outcomes after ^{125}I PRT in uveal melanoma patients [22]. In the case that examination of fellow eyes would reveal deteriorating VA after uveal melanoma treatment, some possible reasons may include unintended radiation exposure, such as stray radiation, indirect radiation-induced nerve damage, inflammation in the affected eye that may cause a similar response in the fellow eye, or treatment-related toxicity from other systemic medications these patients may use. It is important to determine any significant changes in fellow eyes VA or other secondary outcomes occurring for different uveal melanoma treatments, allowing patients to make the best decisions regarding their vision and quality of life after affected eye treatment and frequently progressive loss of vision.

The primary aim of the current study was to examine whether single-eye plaque radiation treatment, when comparing PBT with other forms of uveal melanoma treatment, has any effect on VA of fellow eyes in uveal melanoma patients. Concurrent analysis of fellow eyes was done for changes in IOP up to 10 years post-treatment, and for relationships between different risk factors, i.e., comorbidities, and tumor characteristics, with affected and fellow eyes VA outcomes.

Material and methods

A retrospective chart review was conducted based on electronic medical records of all patients with a diagnosis of uveal melanoma treated by the senior author at Retina Consultants of Alabama in Birmingham, Alabama, from year 2000 till 2022. Exclusion criteria were patients, for whom no treatment was provided during this time period, treatment was completed prior to this time period, and no baseline nor follow-up fellow eyes VA recorded. A total of 213 patients met inclusion criteria, of whom 171 were treated with ^{125}I plaque radiotherapy (PRT), 32 were treated with TTT, and 10 were treated with PBT.

The primary variable, patients' VA at baseline and throughout follow-up time intervals up to 10 years after treatment (when available), was collected. VA measurements in both fellow and affected eyes from follow-up visits at 6 months, 1 year, 2 years, 5 years, and 10 years post-treatment were recorded, when available. Although it was impossible to control how closely each post-operative appointment aligned to the precise follow-up of the chosen intervals, follow-up visits were aligned as closely to their assigned post-operative intervals as possible. Secondary variables, including demographic factors, medical/surgical history, melanoma characteristics (such as size and location at diagnosis), type of treatment received (PRT vs. TTT vs. PBT), and total length of recorded follow-up care from time of treatment, were also recorded.

Data were collected using Excel software (Microsoft Corporation, Redmond, WA), and analyzed with SAS Statistical Software for Windows (SAS Institute Inc., Cary, NC). Paired *t*-tests were employed to compare changes in visual acuity over time (e.g., baseline vs. 6 months) and IOP over time for each treatment type in both fellow and affected eyes. Spearman's rho test was used to determine correlation between melanoma thickness and VA at dif-

ferent follow-up time points. Repeated measures ANOVA was applied to assess interactions between time and comorbidities (i.e., diabetes, hypertension, and coronary artery disease) in VA. *P*-values of ≤ 0.05 (two-sided) were considered statistically significant.

Results

The mean (SD) patient age at the time of uveal melanoma diagnosis was 62.5 (12.7) years. The mean and median length of follow-up was 72 months and 49 months, respectively. Demographically, our patient population included 103 (48.4%) males and 110 (51.6%) females, and was racially homogenous with 212 cases (99.5%) identified as White. Patients in this data set additionally suffered from several systemic diseases mostly associated with ophthalmological disorders, including 118 (55.4%) patients with hypertension, 48 (22.5%) with diabetes mellitus, and 19 (8.9%) with coronary artery disease. The demographics stratified by treatment type are described in Table 1. For tumor characteristics, the median melanoma thickness was 3.9 mm, while 103 (50.7%) of the melanomas were < 4 mm, 67 (33.0%) were 4-6.99 mm, and 33 (16.3%) were 7-11 mm. The median distance from the optic nerve was 6.0 mm, and the median distance from the fovea was 5.0 mm. The melanoma characteristics and treatment types are shown in Table 2. The most common

locations of uveal melanomas were the choroid (86.9%), ciliary body (8.0%), iris (2.3%), and macula (0.9%). In the up to 10-year follow-up period, 19 (9.1%) participants were recorded to have new fellow eye cataract diagnosis.

The baseline VA for the affected and fellow eyes was collected. In Table 3, each baseline VA stratified by treatment type for the affected and fellow eyes, is shown. All fellow eyes started with baseline VA of 20/200 or better. For affected eyes, most started with baseline VA of 20/200 or better, except for six PBT patients having VA values worse than 20/200 at baseline. When sub-categorized by age or sex, there was no significant difference in fellow eye VA outcomes between male and female patients, or between patients of specific age ranges (31-40 years, 41-50 years, etc.) who underwent any of the three treatment modalities examined in this study.

In patients ($n = 171$) treated with PRT, there were no significant changes in fellow eyes VA at 6 months ($p = 0.41$), 1 year ($p = 0.08$), 5 years ($p = 0.45$), or 10 years ($p = 0.49$) post-treatment. It should be noted that the power for each subsequent follow-up interval after baseline decreased, as the number of patients at baseline ($n = 171$) decreased at 6 months ($n = 149$), 1 year ($n = 129$), 2 years ($n = 87$), 5 years ($n = 54$), and 10 years ($n = 28$) post-treatment. At 2 years post-treatment, however, there was a significant increase in the reported fellow eyes visual acuity

Table 1. Demographics stratified by treatment type

Parameter	Plaque radiotherapy with ^{125}I (PRT)	Transpupillary thermotherapy (TTT)	Proton beam therapy (PBT)	Total (n)
Sex, n (%)				
Female	88 (80.0)	17 (15.5)	5 (4.6)	110
Male	83 (80.6)	15 (14.6)	5 (4.9)	103
Race/ethnicity, n (%)				
White	170 (80.2)	32 (15.1)	10 (4.7)	212
African American	1 (100.0)	0	0	1
Other	0	0	0	0
Systemic diseases, n (%)				
Hypertension	96 (81.4)	17 (14.4)	5 (4.2)	118
Coronary artery disease	16 (84.2)	1 (5.3)	2 (10.5)	19
Diabetes mellitus	39 (81.3)	7 (14.6)	2 (4.2)	48

Table 2. Melanoma characteristics stratified by treatment

Parameter	Plaque radiotherapy with ^{125}I (PRT)	Transpupillary thermotherapy (TTT)	Proton beam therapy (PBT)	Total (n or median value)
Melanoma size (mm)				
Number of patients	164	29	9	202
Median (SD)	4.05 (2.3)	3.0 (0.7)	3.9 (1.3)	3.9 (2.2)
Distance from optic nerve (mm)				
Number of patients	155	3	0	158
Median (SD)	6.0 (6.9)	2.0 (7.6)		6.0 (6.9)
Distance from fovea (mm)				
Number of patients	155	3	0	158
Median (SD)	5.0 (6.9)	4.0 (5.5)		5.0 (6.9)

Table 3. Baseline visual acuity (VA) for affected and fellow eyes stratified by treatment

Affected eye, baseline VA	≤ 20/40	20/50-20/200	> 20/200	Total
Plaque radiotherapy with ¹²⁵ I (PRT)	116	48	6	170
Transpupillary thermotherapy (TTT)	20	11	0	31
Proton beam therapy (PBT)	7	3	0	10
Fellow eye, baseline VA	≤ 20/40	20/50-20/200	> 20/200	Total
Plaque radiotherapy with ¹²⁵ I (PRT)	157	14	0	171
Transpupillary thermotherapy (TTT)	29	3	0	32
Proton beam therapy (PBT)	10	0	0	10

($p = 0.04$). Examining the 12 patients (7%) of the PRT group, who developed cataracts in the follow-up period, only 3 (25%) were documented before 2 years post-treatment. In patients ($n = 171$) treated with PRT, there were significant decreases in affected or treated eyes VA at 6 months ($p = 0.0005$), 1 year ($p = 0.0005$), 2 years ($p < 0.0001$), 5 years ($p < 0.0001$), and 10 years ($p = 0.001$) post-treatment.

For the TTT-treated group, there was also no significant change in fellow eyes VA at 6 months ($p = 0.16$), 1 year ($p = 0.74$), 2 years ($p = 0.87$), 5 years ($p = 0.58$), and 10 years ($p = 0.71$) post-treatment. Similar results were observed in affected eyes VA for the TTT sub-group, with no significant VA mean difference compared with baseline at 6 months ($p = 0.15$), 1 year ($p = 0.44$), 2 years ($p = 0.49$), 5 years ($p = 0.58$), and 10 years ($p = 0.17$) post-treatment. Likewise, there was a decreasing sample size power with increased time.

For the PBT patients, there was also no significant change in fellow eyes VA at 6 months ($p = 0.67$), 1 year ($p = 0.68$), 2 years ($p = 0.63$), 5 years ($p = 0.65$), and 10 years ($p = 0.18$) post-treatment. In comparison, for affected eyes, there was a significant difference between 6 months VA ($p = 0.02$) and 1 year VA ($p = 0.01$) compared with baseline VA. There was no significance determined for 2 years ($p = 0.09$) and 5 years (0.05) post-treatment; however, the sample size was already significantly low, with 5 participants at 2 and 5 years, and only 1 participant at 10 years after treatment.

Fellow eyes baseline VA ($n = 202$) was not correlated with melanoma thickness ($\rho = 0.06$, $p = 0.40$). This was the same for fellow eyes VA at 6 months ($\rho = 0.06$, $p = 0.41$), 1 year ($\rho = 0.04$, $p = 0.66$), 2 years ($\rho = -0.06$, $p = 0.54$), 5 years ($\rho = 0.03$, $p = 0.83$), and 10 years ($\rho = 0.20$, $p = 0.57$) post-treatment. The baseline VA of affected eyes ($n = 200$) also did not demonstrate a significant association with melanoma thickness ($\rho = 0.12$, $p = 0.09$).

Furthermore, intraocular pressure (IOP) was examined for each sub-group of patients in this study. The mean IOP for fellow eyes at baseline was 14.87 mmHg (SD, 3.54 mmHg). For fellow eyes, there was no significant change in IOP from baseline regardless of the type of treatment used, i.e., PRT, TTT, or PBT.

The mean IOP for affected eyes at baseline was 14.61 mmHg (SD, 4.05 mmHg). In patients treated with PRT ($n = 171$), affected eyes IOP was significantly increased ($p = 0.008$) at 6-month interval post-treatment, but was not at other follow-up time points. There was no signifi-

cant difference in IOP from baseline IOP values for affected eyes in both the TTT and PBT treatment groups.

A repeated measures ANOVA was conducted to assess the effect of time and each comorbidity (i.e., diabetes, hypertension, or coronary artery disease) on visual acuity. For fellow eyes, there was no significant effect of time, each comorbidity, nor the interaction between time and each comorbidity on visual acuity. There was a significant effect of time on visual acuity for affected eyes in all 3 comorbidity groups (all p values < 0.05). However, there was no significant effect of having any of these 3 comorbidities, nor the interaction between time and each comorbidity on VA.

Discussion

Uveal melanoma requires extensive treatment of the affected eyes, often leading to declining visual acuity and other consequences of tumoricidal treatment. However, we found no significant long-term VA changes in fellow eyes following treatment with either ¹²⁵I PBT, TTT, or PRT. The only exception, a small increase in VA, was observed at 2 years with PRT, which is likely attributable to the low sample size, and being a possible false positive, especially considering no significance for VA at other time points pre- and post-treatment and a likely low contribution from cataract development at this follow-up time. In addition, we found this result to be significant, given that most treated patients had fellow eyes baseline VA values of 20/200 or better (97.2%) and a large subgroup had VA of 20/40 or better (67.8%). This aligns with other findings examining VA changes in fellow eyes from baseline up to 3 years after ¹²⁵I PRT treatment as a secondary outcome [13]. The only other meaningful data following patients up to a decade until the year 2000 for fellow eyes VA comes from the COMS trial comparing ¹²⁵I brachytherapy with enucleation fellow eye outcomes [19]. The authors observed that the mean change in fellow eyes VA was 0.2 (approximately 1 Snellen letter) Snellen lines or less when comparing baseline with each exam point (i.e., 6 months, 1 year, and subsequent year points). This study supports the same finding in fellow eyes VA for ¹²⁵I PRT, while also adding reassurance that TTT and PBT have similar outcomes.

Affected eyes demonstrated a significant change from baseline VA over time for both the ¹²⁵I PRT and PBT treatment groups, in as early as 6 months after treatment. It is well-recognized that affected eyes have poor VA fol-

lowing PRT, with multiple studies sub-categorized by the extent of VA loss in Snellen lines or baseline VA groups [6, 9, 13, 15]. Though PBT is less studied and more limited in treatment availability due to fewer centers offering it worldwide, it has shown the same effect on poor VA when compared baseline with post-treatment time points values [6, 9, 10].

However, TTT did not have any significant changes in affected eyes VA from baseline over every time point up to 10 years. In comparison, TTT has been shown to offer stable visual acuity to patients, with studies reporting preserved VA of 20/80 pre- and post-treatment [23]. This is likely because TTT is a rarely used, older therapy option for small choroidal lesions outside of the macula [1], leading to more chance for a lack of decreased VA. These results can vary due to limited patient selection and tumor localization, but do not discount TTT's ability to still damage the retina. Because the TTT group in this data started with a smaller tumor size (in this case, a median of 3.0 mm with a smaller SD of 0.7 mm compared with PRT with a median of 4.05 mm and a larger SD of 2.3 mm), it is possible that this contributing factor also led to less risk of VA loss in affected eyes.

The analysis of secondary variables, such as tumor thickness and intraocular pressure, provide additional context in the possibility of fellow eye to be affected by a treatment. Fellow eyes VA showed no correlation with melanoma thickness at any time point. This has not been investigated in prior literature, and can be used to reassure patients that despite the size of melanoma thickness, it presents no bearings on contralateral eye having decreased VA. Similar to VA, fellow eyes' IOP showed no significant changes over time from baseline IOP. The COMS trial found similar results, with IOP changing by less than 1 mmHg from baseline to each time point, and the majority of eyes starting with less than 18 mmHg IOP in fellow eyes [19].

Although hypertension has been considered a factor associated with decreased vision after radiotherapy and diabetes with proliferative radiation retinopathy post-plaque therapy [9, 24], we did not find any significant association with each comorbidity and declining VA in affected nor fellow eyes. This is supported by another study on PBT, showing VA values of less than 20/200 at 3 years not associated with diabetes, systemic hypertension, or hypercholesterolemia [25].

A limitation of this study includes non-randomization of patients to treatment groups and uneven sample sizes of each treatment group for outcomes comparison. These uveal melanoma patients had individually varying tumor characteristics, including location and thickness, which influenced treatment selection, but this analysis was restricted to comparing patients based on treatment groups without further stratification of these factors. The assessment of IOP is also limited in meaning because of its lack of correlation with dose, localization, development of conditions, such as neovascular glaucoma, and the use of intravitreal steroid therapies after treatment. Furthermore, the analysis of comorbidities, such as cataract, was focused only on fellow eyes in regard to VA

decline, while further investigation of increased risk of cataract development in fellow eyes was not done due to other confounding factors, such as older age, in this cohort. Moreover, the patients in this study were treated by 1 ophthalmologist, which can affect generalizability of the data. However, many of the findings, especially for affected eyes, align with the results of other, similar studies. Future research can investigate comparable fellow eye findings in other study settings, contrast affected eye findings, and determine any associations with worse visual acuity among different comorbidities for each treatment type.

Conclusions

In summary, our results suggest that fellow eyes may remain stable up to ten years in visual acuity values regardless of treatment type, i.e., ¹²⁵I PBT, TTT, or PRT. Some risk factors, which show an impact on VA of affected eyes over time, such as tumor thickness and the presence of comorbidities, i.e., hypertension or diabetes, did not have any impact on fellow eye VA outcomes. This can provide reassurance to patients regarding risks in ocular health as well as ability to rely on the vision of untreated contralateral eyes up to a decade, when choosing treatment approaches for uveal melanoma.

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Disclosures

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